



Name \_\_\_\_\_

zID e.g. z1234567

Z \_\_\_\_\_

Fill zID digit bubbles:

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|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 |

**Q1.** In fieldwork you often tape the circumference of a tree at breast height but need diameter (DBH) to use a height-DBH model. What is the correct conversion from measured circumference  $C$  to diameter  $d$ ?

- a.  $d = C^2 / \pi$
- b.  $d = \sqrt{C/\pi}$
- c.  $d = C / \pi$
- d.  $d = \pi C$
- e.  $d = 2C / \pi$

**Answer Q1** (A) (B) (C) (D) (E)

**Q2.** You are comparing two models that predict a vegetation variable  $Y$  from a sensor index  $X$ . Your priority is to minimise absolute prediction error in interpretable units of  $Y$ . Which metric should guide your choice?

- a. Root Mean Squared Error (RMSE)
- b. p-value of the slope
- c. Adjusted  $R^2$
- d. Correlation coefficient  $r$
- e.  $R^2$

**Answer Q2** (A) (B) (C) (D) (E)

**Q3.** You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
- b. Map crowns in an equal-area projection and use crown area as height proxies.
- c. Fit a statistical model of height from DBH using a training set, then use it to predict heights for the remaining trees.
- d. Measure a few heights, take the mean, and assign it to all trees.
- e. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.

**Answer Q3** (A) (B) (C) (D) (E)

**Q4.** Why did the lecture emphasize using simple formulae carefully before moving into software tools?

- a. Coding removes the need to think about units or measurement.
- b. Formulae are only relevant for drawing figures, not analysis.
- c. Software can only run if every formula is typed from memory.
- d. Environmental science avoids mathematical calculations once data are collected.
- e. Quantitative interpretation still depends on understanding what measurements and formulae represent.

**Answer Q4** (A) (B) (C) (D) (E)



**Q5.** In the lecture, what was the purpose of discussing RStudio setup before the labs?

- a. To discourage students from using local computers.
- b. To demonstrate that coding is separate from environmental science.
- c. To make sure students could use the required computing environment for practical work.
- d. To assess students before they had seen the course material.
- e. To replace all later lectures with self-guided programming tasks.

**Answer Q5** (A) (B) (C) (D) (E)

**Q6.** Which statement best describes the role of a cloud workaround such as Google Colab in the lecture?

- a. It is used only for watching lecture videos.
- b. It is required before students can learn RStudio.
- c. It is the default tool for all students regardless of their computer setup.
- d. It automatically calculates all answers without coding.
- e. It is a backup option for students who cannot run the local software environment.

**Answer Q6** (A) (B) (C) (D) (E)

